



Cisco Nexus 9336C-FX2 ACI-Mode Switch Hardware Installation Guide

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CONTENTS

PREFACE

Preface **vii**

Audience **vii**

Documentation Conventions **vii**

Documentation Feedback **viii**

Obtaining Documentation and Submitting a Service Request **viii**

CHAPTER 1

Overview **1**

Overview **1**

CHAPTER 2

Preparing the Site **5**

Temperature Requirements **5**

Humidity Requirements **5**

Altitude Requirements **5**

Dust and Particulate Requirements **6**

Minimizing Electromagnetic and Radio Frequency Interference **6**

Shock and Vibration Requirements **7**

Grounding Requirements **7**

Planning for Power Requirements **7**

Airflow Requirements **8**

Rack and Cabinet Requirements **9**

Clearance Requirements **10**

CHAPTER 3

Installing the Switch Chassis **11**

Safety **11**

Installation Options with Racks and Cabinets **12**

Airflow Considerations **12**

Installation Guidelines	12
Unpacking and Inspecting the Switch	13
Installing the Switch	14
Grounding the Chassis	18
Starting the Switch	20

CHAPTER 4	Connecting the Switch to the ACI Fabric	23
	ACI Fabric Topology	23
	Preparing to Connect to Other Devices	24
	Connecting Leaf Switches to APICs	25
	Connecting Leaf Switches to Spine Switches	26
	Setting Up an Optional Console Interface	29
	Setting Up an Optional Management Connection	30
	Maintaining Transceivers and Optical Cables	30

CHAPTER 5	Replacing Components	33
	Replacing a Fan Module	33
	Removing a Fan Module	33
	Installing a Fan Module	34
	Replacing a Power Supply Module	35
	Replacing an AC Power Supply	35
	Replacing a High Voltage (HVAC/HVDC) Power Supply	36
	Replacing a DC Power Supply	38

APPENDIX A	Rack Specifications	41
	Overview of Racks	41
	General Requirements for Cabinets and Racks	41
	Requirements Specific to Standard Open Racks	42
	Requirements Specific to Perforated Cabinets	42
	Cable Management Guidelines	42

APPENDIX B	System Specifications	43
	Environmental Specifications	43
	Switch Dimensions	43

Switch and Module Weights and Quantities	44
Transceiver and Cable Specifications	44
Switch Power Input Requirements	44
Power Specifications	45
1100-W AC Power Supply Specifications	45
Power Cable Specifications	46
Power Cable Specifications for AC Power Supplies	46
HVAC/HVDC Power Cables Supported by ACI-Mode and NX-OS Mode Switches	47
Regulatory Standards Compliance Specifications	47

APPENDIX C

LEDs	49
Switch Chassis LEDs	49
Lane Link LEDs	50
Fan Module LEDs	50
Power Supply LEDs	50

APPENDIX D

Additional Kits	53
Accessory Kit	53

APPENDIX E

Site Preparation and Maintenance Records	55
Site Preparation Checklist	55
Contact and Site Information	56
Chassis and Module Information	57



Preface

- [Audience, on page vii](#)
- [Documentation Conventions, on page vii](#)
- [Documentation Feedback, on page viii](#)
- [Obtaining Documentation and Submitting a Service Request, on page viii](#)

Audience

This publication is for network administrators who install, configure, and maintain Cisco Nexus switches.

Documentation Conventions

Command descriptions use the following conventions:

Convention	Description
bold	Bold text indicates the commands and keywords that you enter literally as shown.
<i>Italic</i>	Italic text indicates arguments for which the user supplies the values.
[x]	Square brackets enclose an optional element (keyword or argument).
[x y]	Square brackets enclosing keywords or arguments separated by a vertical bar indicate an optional choice.
{x y}	Braces enclosing keywords or arguments separated by a vertical bar indicate a required choice.
[x {y z}]	Nested set of square brackets or braces indicate optional or required choices within optional or required elements. Braces and a vertical bar within square brackets indicate a required choice within an optional element.
<i>variable</i>	Indicates a variable for which you supply values, in context where italics cannot be used.

Convention	Description
string	A nonquoted set of characters. Do not use quotation marks around the string or the string will include the quotation marks.

Examples use the following conventions:

Convention	Description
<code>screen font</code>	Terminal sessions and information the switch displays are in screen font.
boldface screen font	Information you must enter is in boldface screen font.
<i>italic screen font</i>	Arguments for which you supply values are in italic screen font.
< >	Nonprinting characters, such as passwords, are in angle brackets.
[]	Default responses to system prompts are in square brackets.
!, #	An exclamation point (!) or a pound sign (#) at the beginning of a line of code indicates a comment line.

Documentation Feedback

To provide technical feedback on this document, or to report an error or omission, please send your comments to . We appreciate your feedback.

Obtaining Documentation and Submitting a Service Request

For information on obtaining documentation, using the Cisco Bug Search Tool (BST), submitting a service request, and gathering additional information, see *What's New in Cisco Product Documentation* at: <http://www.cisco.com/c/en/us/td/docs/general/whatsnew/whatsnew.html>

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CHAPTER 1

Overview

- [Overview, on page 1](#)

Overview

The Cisco Nexus 9336C-FX2 switch (N9K-C9336C-FX2) is a 1-RU, fixed-port switch designed for spine-leaf-APIC deployment in data centers. This switch has 36 40/100-Gigabit QSFP28 ports. Ports 31-36 support uplink (2 ports needed for minimum uplink, use ports 35-36). All ports support 10-Gigabit with a QSA adapter (CVR-QSF-SFP10G). Ports 1-30 support 40/100-Gigabit Breakout.

To determine which transceivers, adapters, and cables this switch supports, see the [Cisco Transceiver Modules Compatibility Information](#) document.

This switch includes the following user-replaceable components:

- Fan modules (three) with the following airflow choices:
 - Port-side exhaust airflow with blue coloring (NXA-FAN-65CFM-PE)
 - Port-side intake airflow with burgundy coloring (NXA-FAN-65CFM-PI)
- Power supply modules (two—one for operations and one for redundancy [1+1]) with the following choices:
 - 1100-W AC power supply with port-side exhaust airflow (blue coloring) (NXA-PAC-1100W-PE2)
 - 1100-W AC power supply with port-side intake airflow (burgundy coloring) (NXA-PAC-1100W-PI2)
 - 1100-W HVAC/HVDC power supply with port-side exhaust airflow (blue coloring) (NXA-PHV-1100W-PE)
 - 1100-W HVAC/HVDC power supply with port-side intake airflow (burgundy coloring) (NXA-PHV-1100W-PI)



Note

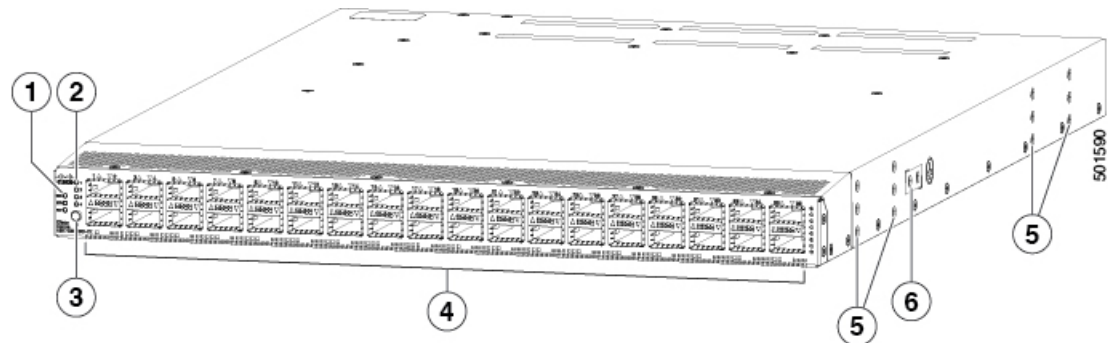
Both power supplies should use the same type of power source. Do not mix AC and DC power sources.



Note All fan modules and power supplies must use the same airflow direction during operations.

The following figure shows the hardware features seen from the port side of the chassis.

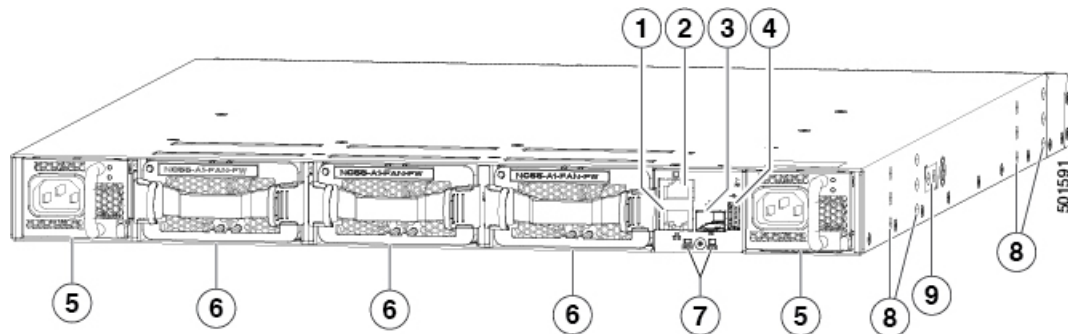
Figure 1: Port Side View of the Cisco Nexus 9336C-FX2 Switch



1	Chassis LEDs (Beacon [BCN], Status [STS], and Environment [ENV])	4	36 ports that can be configured differently for APIC and spine switch connections
2	Lane selector LEDs	5	Screw holes (6) for attaching rack mounting brackets.
3	Lane selection button	6	Screw holes (2) for attaching grounding lug.

The following figure shows the hardware features seen from the power supply side of the chassis.

Figure 2: Power Supply Side View of the Cisco Nexus 9336C-FX2 Switch



1	Out-of-band management port (SFP port)	6	Three fan modules with fan slot 1 on the left and fan slot 3 on the right
2	Console port (RS232 port)	7	Chassis LEDs (Beacon [BCN] and Status [STS])

3	Out-of-band management port (RJ-45 port)	8	Screw holes (6) for attaching rack mounting brackets
4	USB port used for saving or copying functions	9	Screw holes (2) for attaching grounding lug.
5	Two power supplies (one used for operations and one used for redundancy) (AC power supplies shown) with power supply slot 1 on the left and slot 2 on the right		



Note USB support is limited to USB 2.0 devices that use less than 2.5 W (less than 0.5 A inclusive of surge current). Devices, such as external hard drives, that instantaneously draw more than 0.5 A are not supported.

Depending on whether you plan to position the ports in a hot or cold aisle, you can order the fan and power supply modules with port-side intake (burgundy colored) or port-side exhaust (blue colored) airflow. All of the power supply and fan modules must have the same coloring.

The fan and power supply modules are field replaceable and you can replace one fan module or one power supply module during operations so long as the other modules are installed and operating. If you have only one power supply installed, you can install the replacement power supply in the open slot before removing the original power supply.



Note All of the fan and power supply modules must have the same direction of airflow. Otherwise, the switch can overheat and shut down.



Caution If the switch has port-side intake airflow (burgundy coloring for fan modules), you must locate the ports in the cold aisle. If the switch has port-side exhaust airflow (blue coloring for fan modules), you must locate the ports in the hot aisle. If you locate the air intake in a hot aisle, the switch can overheat and shut down.



CHAPTER 2

Preparing the Site

- [Temperature Requirements, on page 5](#)
- [Humidity Requirements, on page 5](#)
- [Altitude Requirements, on page 5](#)
- [Dust and Particulate Requirements, on page 6](#)
- [Minimizing Electromagnetic and Radio Frequency Interference, on page 6](#)
- [Shock and Vibration Requirements, on page 7](#)
- [Grounding Requirements, on page 7](#)
- [Planning for Power Requirements, on page 7](#)
- [Airflow Requirements, on page 8](#)
- [Rack and Cabinet Requirements, on page 9](#)
- [Clearance Requirements, on page 10](#)

Temperature Requirements

The switch requires an operating temperature of 32 to 104 degrees Fahrenheit (0 to 40 degrees Celsius). If the switch is not operating, the temperature must be between –40 to 158 degrees Fahrenheit (–40 to 70 degrees Celsius).

Humidity Requirements

High humidity can cause moisture to enter the switch. Moisture can cause corrosion of internal components and degradation of properties such as electrical resistance, thermal conductivity, physical strength, and size. The switch is rated to withstand from 5- to 95-percent (non-condensing) relative humidity.

Buildings in which the climate is controlled by air-conditioning in the warmer months and by heat during the colder months usually maintain an acceptable level of humidity for the switch equipment. However, if the switch is located in an unusually humid location, you should use a dehumidifier to maintain the humidity within an acceptable range.

Altitude Requirements

The following table lists the maximum altitude that this switch is tested to operate. This switch is rated to operate at altitudes from 0 to 10,000 feet (0 to 3,048 meters). If you operate this switch at a higher altitude

(low pressure), the efficiency of forced and convection cooling is reduced and can result in electrical problems that are related to arcing and corona effects. This condition can also cause sealed components with internal pressure, such as electrolytic capacitors, to fail or to perform at a reduced efficiency.

Dust and Particulate Requirements

Exhaust fans cool power supplies and system fans cool switches by drawing in air and exhausting air out through various openings in the chassis. However, fans also ingest dust and other particles, causing contaminant buildup in the switch and increased internal chassis temperature. A clean operating environment can greatly reduce the negative effects of dust and other particles, which act as insulators and interfere with the mechanical components in the switch.

In addition to regular cleaning, follow these precautions to avoid contamination of your switch:

- Do not permit smoking near the switch.
- Do not permit food or drink near the switch.

Minimizing Electromagnetic and Radio Frequency Interference

Electromagnetic interference (EMI) and radio frequency interference (RFI) from the switch can adversely affect other devices, such as radio and television (TV) receivers, operating near the switch. Radio frequencies that emanate from the switch can also interfere with cordless and low-power telephones. Conversely, RFI from high-power telephones can cause spurious characters to appear on the switch monitor.

RFI is defined as any EMI with a frequency above 10 kHz. This type of interference can travel from the switch to other devices through the power cable and power source or through the air as transmitted radio waves. The Federal Communications Commission (FCC) publishes specific regulations to limit the amount of EMI and RFI that can be emitted by computing equipment. Each switch meets these FCC regulations.

To reduce the possibility of EMI and RFI, follow these guidelines:

- Cover all open expansion slots with a blank filler plate.
- Always use shielded cables with metal connector shells for attaching peripherals to the switch.

When wires are run for any significant distance in an electromagnetic field, interference can occur between the field and the signals on the wires with the following implications:

- Bad wiring can result in radio interference emanating from the plant wiring.
- Strong EMI, especially when it is caused by lightning or radio transmitters, can destroy the signal drivers and receivers in the chassis and even create an electrical hazard by conducting power surges through lines into equipment.



Note

To predict and prevent strong EMI, you might need to consult experts in radio frequency interference (RFI).

The wiring is unlikely to emit radio interference if you use twisted-pair cable with a good distribution of grounding conductors. If you exceed the recommended distances, use a high-quality twisted-pair cable with one ground conductor for each data signal when applicable.

**Caution**

If the wires exceed the recommended distances, or if wires pass between buildings, give special consideration to the effect of a lightning strike in your vicinity. The electromagnetic pulse caused by lightning or other high-energy phenomena can easily couple enough energy into unshielded conductors to destroy electronic switches. You might want to consult experts in electrical surge suppression and shielding if you had similar problems in the past.

Shock and Vibration Requirements

The switch has been shock- and vibration-tested for operating ranges, handling, and earthquake standards.

Grounding Requirements

The switch is sensitive to variations in voltage supplied by the power sources. Overvoltage, undervoltage, and transients (or spikes) can erase data from memory or cause components to fail. To protect against these types of problems, ensure that there is an earth-ground connection for the switch. You can connect the grounding pad on the switch either directly to the earth-ground connection or to a fully bonded and grounded rack.

When you properly install the chassis in a grounded rack, the switch is grounded because it has a metal-to-metal connection to the rack. Alternatively, you can ground the chassis by using a customer-supplied grounding cable that meets your local and national installation requirements (we recommend 6-AWG wire for U.S. installations) connected to the chassis with a grounding lug (provided in the switch accessory kit) and to the facility ground.

**Note**

You automatically ground AC power supplies when you connect them to AC power sources. For DC power supplies, you must connect a grounding wire when wiring the power supply to the DC power source.

Planning for Power Requirements

The switch includes two power supplies (1-to-1 redundancy with current sharing) in one of the following combinations:

- Two 1100-W AC power supplies
- Two 1100-W HVAC/HVDC power supplies

**Note**

Both power supplies must be the same type. Do not mix AC and DC power supplies in the same chassis.

**Note**

For $n+1$ redundancy, you can use one or two power sources for the two power supplies. For $n+n$ redundancy, you must use two power sources and connect each power supply to a separate power source.

The power supplies are rated to output up to 1100 W, but the switch requires less than those amounts of power from the power supply. To operate the switch you must provision enough power from the power source to cover the requirements of both the switch and a power supply. Typically, this switch and a power supply require about 367 W of power input from the power source, but you must provision as much as 777 W power input from the power source to cover peak demand.

**Note**

Some of the power supply modules have Underwriter Labs (UL) rating capabilities that exceed the switch requirements. When calculating power requirements, use the switch requirements to determine the amount of power required for the power supplies.

To minimize the possibility of circuit failure, make sure that each power-source circuit used by the switch is dedicated to the switch.

**Note**

For AC input application, please refer to the statement below:

**Warning****Statement 1005—Circuit Breaker**

This product relies on the building's installation for short-circuit (overcurrent) protection. Ensure that the protective devices are rated not greater than 20A (North America), 16A (Europe), and 13A (UK).

**Note**

For DC input application, please refer to the statement below:

**Warning****Statement 1005—Circuit Breaker**

This product relies on the building's installation for short-circuit (overcurrent) protection.

- Ensure that the protective devices are rated not greater than 40A when the switch is powered with regular DC power supplies (rated 48-60VDC).
- Ensure that the protective devices are rated not greater than 10A when the switch is powered with HVDC power supplies (rated 240-350VDC).

**Note**

For the power cables to use with the power supplies, see [Power Cable Specifications, on page 46](#).

Airflow Requirements

The switch is designed to be positioned with its ports in either the front or the rear of the rack depending on your cabling and maintenance requirements. Depending on which side of the switch faces the cold aisle, you

must have fan and power supply modules that move the coolant air from the cold aisle to the hot aisle in one of the following ways:

- Port-side exhaust airflow—Coolant air enters the chassis through the fan and power supply modules in the cold aisle and exhausts through the port end of the chassis in the hot aisle.
- Port-side intake airflow—Coolant air enters the chassis through the port end in the cold aisle and exhausts through the fan and power supply modules in the hot aisle.

You can identify the airflow direction of each fan and power supply module by its coloring as follows:

- Blue coloring indicates port-side exhaust airflow.
- Burgundy coloring indicates port-side intake airflow.

**Note**

To prevent the switch from overheating and shutting down, you must position the air intake for the switch in a cold aisle, and all of the fan and power supply modules must have the same direction of airflow (even if their coloring is different). If you must change the airflow direction for the switch, you must shutdown the switch before changing the modules.

Rack and Cabinet Requirements

You can install the following types of racks or cabinets for your switch:

- Standard perforated cabinets
- Solid-walled cabinets with a roof fan tray (bottom-to-top cooling)
- Standard open four-post Telco racks

Work with your cabinet vendors to determine which of their cabinets meet the following requirements or see the Cisco Technical Assistance Center (TAC) for recommendations:

- Use a standard 19-inch (48.3-cm), four-post Electronic Industries Alliance (EIA) cabinet or rack with mounting rails that conform to English universal hole spacing per section 1 of the ANSI/EIA-310-D-1992 standard.
- The depth of a four-post rack must be 24 to 32 inches (61.0 to 81.3 cm) between the front and rear mounting rails (for proper mounting of the bottom-support brackets or other mounting hardware).

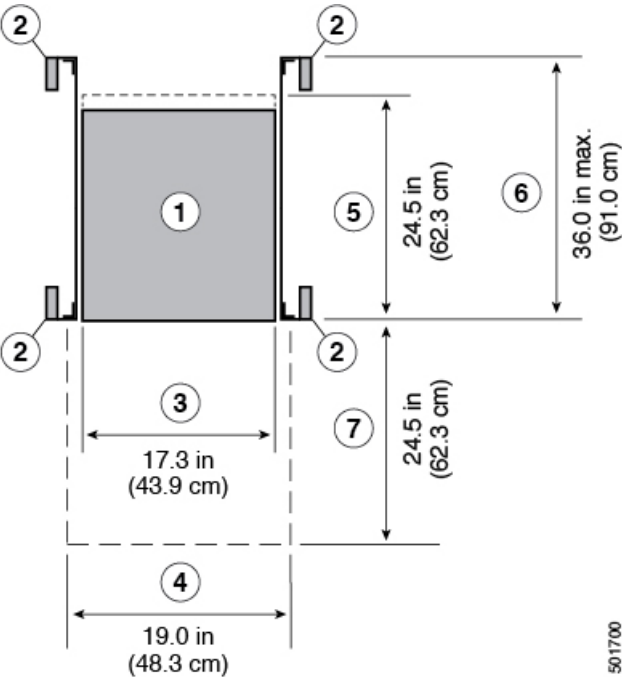
Additionally, you must have power receptacles located within reach of the power cords used with the switch.

**Warning****Statement 1048—Rack Stabilization**

Stability hazard. The rack stabilizing mechanism must be in place, or the rack must be bolted to the floor before you slide the unit out for servicing. Failure to stabilize the rack can cause the rack to tip over.

Clearance Requirements

You must provide the chassis with adequate clearance between the chassis and any other rack, device, or structure so that you can properly install the chassis, route cables, provide airflow, and maintain the switch. For the clearances required for an installation of this chassis in a four-post rack, see the following figure.



1	Chassis	5	Depth of the chassis
2	Vertical rack-mount posts and rails	6	Maximum extension of the bottom-support rails
3	Chassis width	7	Depth of the front clearance area (this equals the depth of the chassis)
4	Width of the front clearance area (this equals the width of the chassis with two rack-mount brackets attached to it)		



Note Both the front and rear of the chassis must be open to both aisles for airflow.



CHAPTER 3

Installing the Switch Chassis

- [Safety, on page 11](#)
- [Installation Options with Racks and Cabinets, on page 12](#)
- [Airflow Considerations, on page 12](#)
- [Installation Guidelines, on page 12](#)
- [Unpacking and Inspecting the Switch, on page 13](#)
- [Installing the Switch, on page 14](#)
- [Grounding the Chassis, on page 18](#)
- [Starting the Switch, on page 20](#)

Safety

Before you install, operate, or service the switch, see the *Regulatory, Compliance, and Safety Information for the Cisco Nexus 3000 and 9000 Series* for important Safety Information.



Warning

Statement 1071—Warning Definition

IMPORTANT SAFETY INSTRUCTIONS

This warning symbol means danger. You are in a situation that could cause bodily injury. Before you work on any equipment, be aware of the hazards involved with electrical circuitry and be familiar with standard practices for preventing accidents. Use the statement number provided at the end of each warning to locate its translation in the translated safety warnings that accompanied this device.

SAVE THESE INSTRUCTIONS



Warning

Statement 1017—Restricted Area

This unit is intended for installation in restricted access areas. A restricted access area can be accessed only through the use of a special tool, lock and key, or other means of security.

**Warning****Statement 1030—Equipment Installation**

Only trained and qualified personnel should be allowed to install, replace, or service this equipment.

Installation Options with Racks and Cabinets

You can install the switch in the following types of racks using the rack-mount kit shipped with the switch:

- Open EIA rack
- Perforated EIA cabinet

The rack or cabinet that you use must meet the requirements listed in [General Requirements for Cabinets and Racks, on page 41](#).

The rack-mount kit enables you to install the switch into racks of varying depths. You can use the rack-mount kit parts to position the switch with easy access to either the port connections end of the chassis or the end of the chassis with the fan and power supply modules. For instructions on how to install the rack-mount kit, see the [Installing the Switch, on page 14](#).

Airflow Considerations

The switch comes with fan and power supply modules that have either port-side intake or port-side exhaust airflow for cooling the switch. If you are positioning the port end of the switch in a cold aisle, make sure that the switch has port-side intake fan modules with burgundy coloring. If you are positioning the fan and power supply modules in a cold aisle, make sure that the switch has port-side exhaust fan modules with blue colorings. All fan modules must have the same direction of airflow.

Installation Guidelines

When installing the switch, follow these guidelines:

- Record equipment and installation information in the forms presented in [Chassis and Module Information](#) as you install and configure the switch.
- Ensure that there is adequate clearance space around the switch to allow for servicing the switch and for adequate airflow.
- Ensure that the chassis can be adequately grounded. If the switch is not mounted in a grounded rack, we recommend connecting both the system ground on the chassis directly to an earth ground.
- Ensure that the site power meets the power requirements for the switch. If available, you can use an uninterruptible power supply (UPS) to protect against power failures.

**Caution**

Avoid UPS types that use ferroresonant technology. These UPS types can become unstable with the switch, which can have substantial current draw fluctuations because of fluctuating data traffic patterns.

- Ensure that circuits are sized according to local and national codes. Typically, this often requires one or both of the following:
 - AC power supplies typically require at least a 15-A or 20-A AC circuit, 100 to 240 VAC, and a frequency of 50 to 60 Hz.
 - HVAC/HVDC power supplies require the following:
 - HVAC input voltage range of 100 to 277 VAC with a frequency of 50 to 60 Hz
 - HVDC input voltage range of -240 to -380 VDC

**Caution**

To prevent loss of input power, ensure the total maximum loads on the circuits supplying power to the switch are within the current ratings for the wiring and breakers.

Unpacking and Inspecting the Switch

Before you install the switch, be sure to unpack and inspect the switch for damage or missing components. If anything is missing or damaged, contact your customer service representative immediately.



Tip Keep the shipping container in case the chassis requires shipping at a later time.

Before you begin

Before you unpack the switch and before you handle any switch components, be sure that you are wearing a grounded electrostatic discharge (ESD) strap. To ground the strap, attach it directly to an earth ground or to a grounded rack or grounded chassis (there must be a metal-to-metal connection to the earth ground).

Step 1 Compare the shipment to the equipment list provided by your customer service representative and verify that you have received all items, including the following:

- Accessory Kit

Note For the contents of these kits, see the [Additional Kits](#).

Step 2 Check for damage and report any discrepancies or damage to your customer service representative. Have the following information ready:

- Invoice number of shipper (see packing slip)

- Model and serial number of the damaged unit
- Description of damage
- Effect of damage on the installation

Step 3 Check to be sure that each of the power supply and the fan tray modules have the expected direction of airflow as follows:

- Port-side intake airflow modules
 - Burgundy (fan modules and power supplies)
- Port-side exhaust airflow modules
 - Blue (fan modules and power supplies)

Note All power supplies and fan modules must have the same direction of airflow.

Installing the Switch

To install the switch, you must attach front and rear mounting brackets to the switch, install slider rails on the rear of the rack, slide the switch onto the slider rails, and secure the switch to the front of the rack. Typically, the front of the rack is the side easiest to access for maintenance.



Note You must supply the eight 10-32 or 12-24 screws required to mount the slider rails and switch to the rack.

Before you begin

- You have inspected the switch shipment to ensure that you have everything ordered.
 - Make sure that the switch rack-mount kit includes the following parts:
 - Front rack-mount brackets (2)
 - Rear rack-mount brackets (2)
 - Slider rails (2)
 - M4 x 0.7 x 8-mm Phillips countersink screws (12)
 - The rack is installed and secured to its location.
-

Step 1 Install two front-mount brackets to the switch as follows:

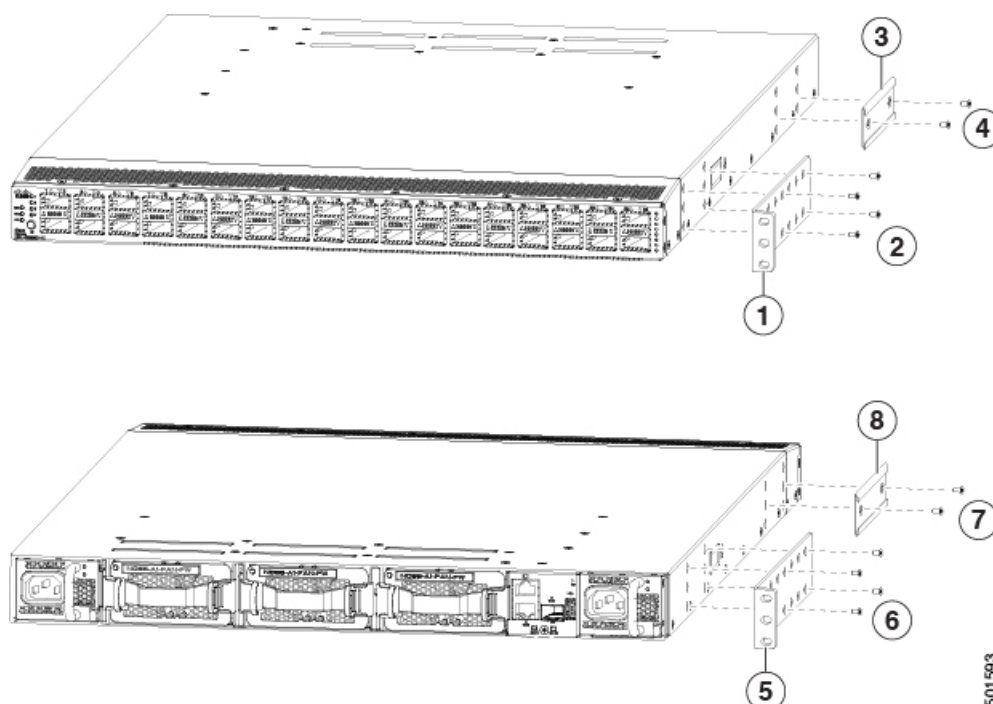
a) Determine which end of the chassis is to be located in the cold aisle as follows:

- If the switch has port-side intake modules (fan modules with burgundy coloring), position the switch so that its ports will be in the cold aisle.

- If the switch has port-side exhaust modules (fan modules with blue coloring), position the switch so that its fan and power supply modules will be in the cold aisle.
- b) Position a front-mount bracket so that four of its screw holes are aligned to the screw holes on the side of the chassis.

Note You can align any four of the holes in the front rack-mount bracket to four of the six screw holes on the side of the chassis (see the two ways to mount these brackets on a typical chassis, in following figure). The holes that you use depend on the requirements of your rack and the amount of clearance required for interface cables (3 inches [7.6 mm] minimum) and module handles (1 inch [2.5 mm] minimum).

Figure 3: Attaching Mounting Brackets



1	Front rack-mount bracket aligned to the port end of the chassis	5	Front rack-mount bracket aligned to the module end of the chassis
2	Four M4 screws used to attach the bracket to the chassis	6	Four M4 screws used to attach the bracket to the chassis
3	Rear rack-mount guide aligned to the module end of the chassis	7	Two M4 screws used to attach the bracket to the chassis
4	Two M4 screws used to attach the bracket to the chassis	8	Rear rack-mount guide aligned to the port end of the chassis

- c) Secure the front-mount bracket to the chassis using four M4 screws and tighten each screw to 12 in-lb (1.36 N·m) of torque.
- d) Repeat Step 1 for the other front rack-mount bracket on the other side of the switch and be sure to position that bracket the same distance from the front of the switch.

Step 2 Install the two rear rack-mount brackets on the chassis as follows:

- a) Align the two screw holes on a rear rack-mount bracket to the middle two screw holes in the remaining six screw holes on a side of the chassis. If you are aligning the guide to holes that are near the port connections end of the chassis, see Callout 3 in the previous figure. Otherwise, see Callout 7 in the previous figure.
- b) Attach the guide to the chassis using two M4 screws (see Callout 4 or 8 in the previous figure). Tighten the screws to 12 in-lb (1.36 N·m) of torque.
- c) Repeat Step 2 for the other rear rack-mount bracket on the other side of the switch.

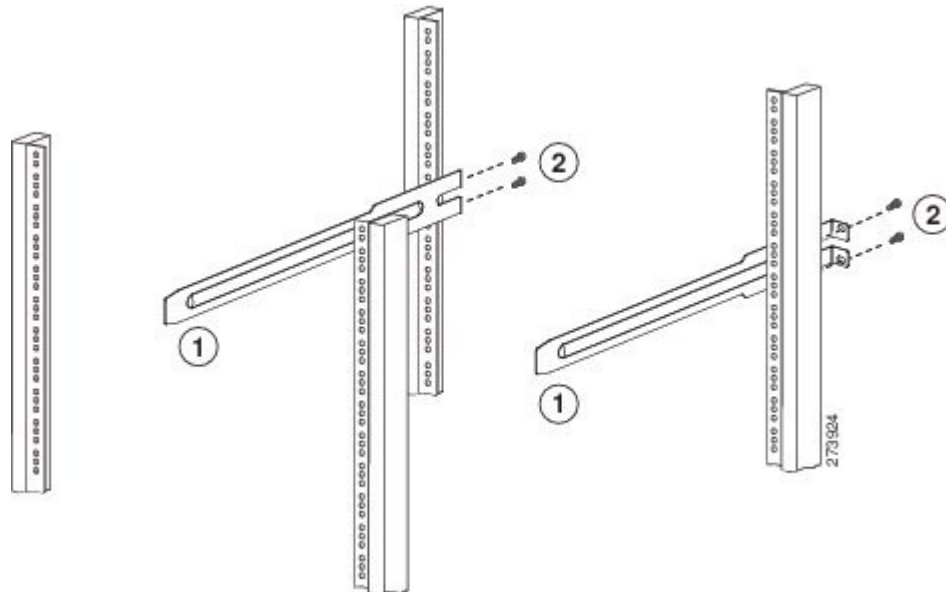
Step 3

If you are not installing the chassis into a grounded rack, you must attach a customer-supplied grounding wire to the chassis as explained in [Grounding the Chassis, on page 18](#). If you are installing the chassis into a grounded rack, you can skip this step.

Step 4

Install the slider rails on the rack or cabinet as follows:

- a) Determine which two posts of the rack or cabinet you should use for the slider rails. Of the four vertical posts in the rack or cabinet, two will be used for the front mount brackets attached to the easiest accessed end of the chassis, and the other two posts will have the slider rails.
- b) Position a slider rail at the desired level on the back side of the rack and use two 12-24 screws or two 10-32 screws, depending on the rack thread type, to attach the rails to the rack (see the following figure). Tighten 12-24 screws to 30 in-lb (3.39 N·m) of torque and tighten 10-32 screws to 20 in-lb (2.26 N·m) of torque.



1	Slider rail with screw holes aligned to screw holes in rack	2	Two customer-supplied 12-24 or 10-32 screws used to attach each slider rail to the rack
---	---	---	---

- c) Repeat Step 3 to attach the other slider rail to the other side of the rack.

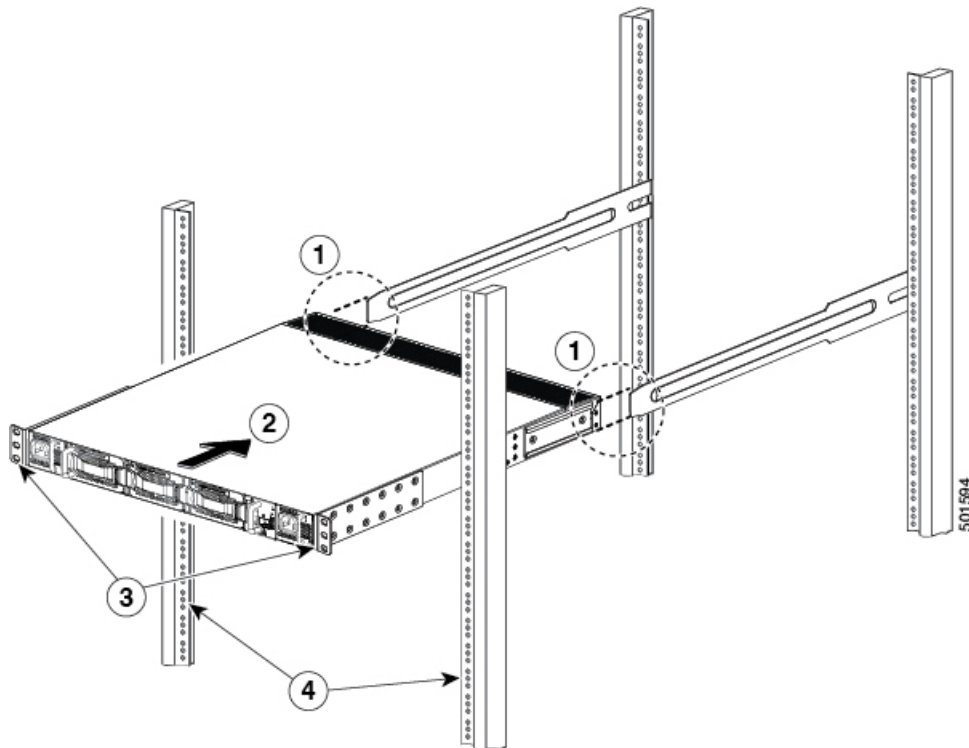
To make sure that the slider rails are at the same level, you should use a level tool, tape measure, or carefully count the screw holes in the vertical mounting rails.

Step 5

Insert the switch into the rack and attach it as follows:

- a) Holding the switch with both hands, position the two rear rack-mount brackets on the switch between the rack or cabinet posts that do not have slider rails attached to them (see the following figure).

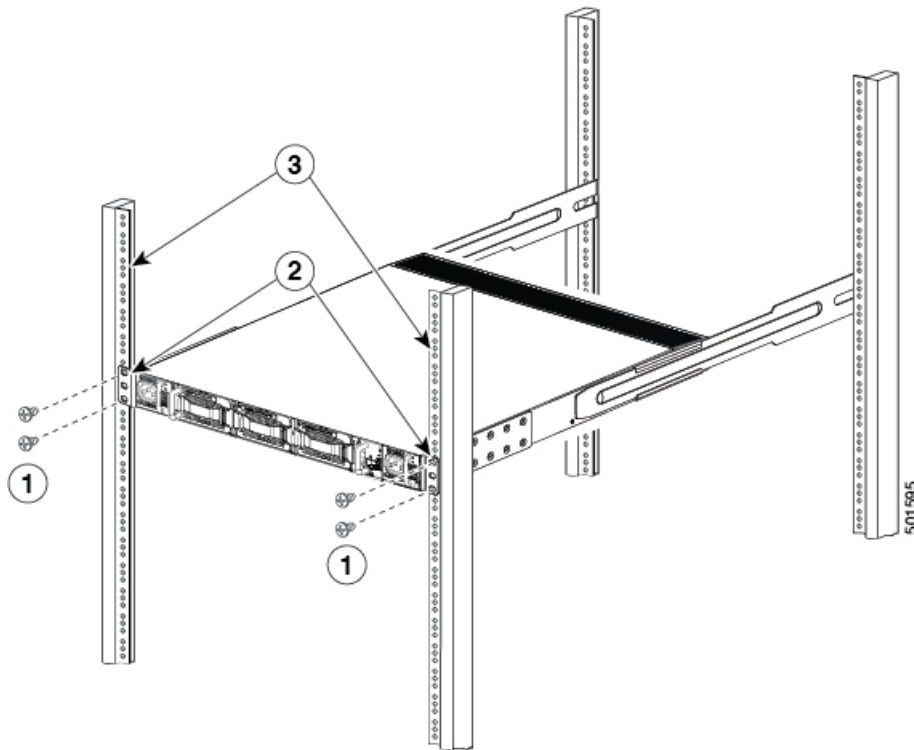
Figure 4: Sliding the Switch onto the Mounting Rails



1	Align the two rear rack-mount bracket guides with the slider rails installed in the rack.	3	Front-mount brackets.
2	Slide the rack-mount guides onto the slider rails until the front rack-mount brackets come in contact with the front rack-mount rails.	4	Mounting rails on rack or cabinet posts.

- b) Align the two rear rack-mount guides on either side of the switch with the slider rails installed in the rack. Slide the rack-mount guides onto the slider rails, and then gently slide the switch all the way into the rack until the front rack-mount brackets come in contact with two rack or cabinet posts.
- c) Holding the chassis level, insert two screws (12-24 or 10-32, depending on the rack type) in each of the two front rack-mount brackets (using a total of four screws) and into the cage nuts or threaded holes in the vertical rack-mounting rails (see the following figure).

Figure 5: Securing the Switch to the Rack



1	Fasten the chassis to the front of the rack with two 12-24 or 10-32 screws on each side.	3	Mounting rails on rack or cabinet posts.
2	Front-mount bracket.		

d) Tighten the 10-32 screws to 20 in-lb (2.26 N·m) or tighten the 12-24 screws to 30 in-lb (3.39 N·m).

Step 6 If you attached a grounding wire to the chassis grounding pad, connect the other end of the wire to the facility ground.

Grounding the Chassis

The switch chassis is automatically grounded when you properly install the switch in a grounded rack with metal-to-metal connections between the switch and rack.

You can alternatively ground the chassis (this is required if the rack is not grounded) by attaching a customer-supplied grounding cable to the chassis grounding pad and the facility ground.



Warning Statement 1024—Ground Conductor

This equipment must be grounded. Never defeat the ground conductor or operate the equipment in the absence of a suitably installed ground conductor. Contact the appropriate electrical inspection authority or an electrician if you are uncertain that suitable grounding is available.


Warning Statement 1046—Installing or Replacing the Unit

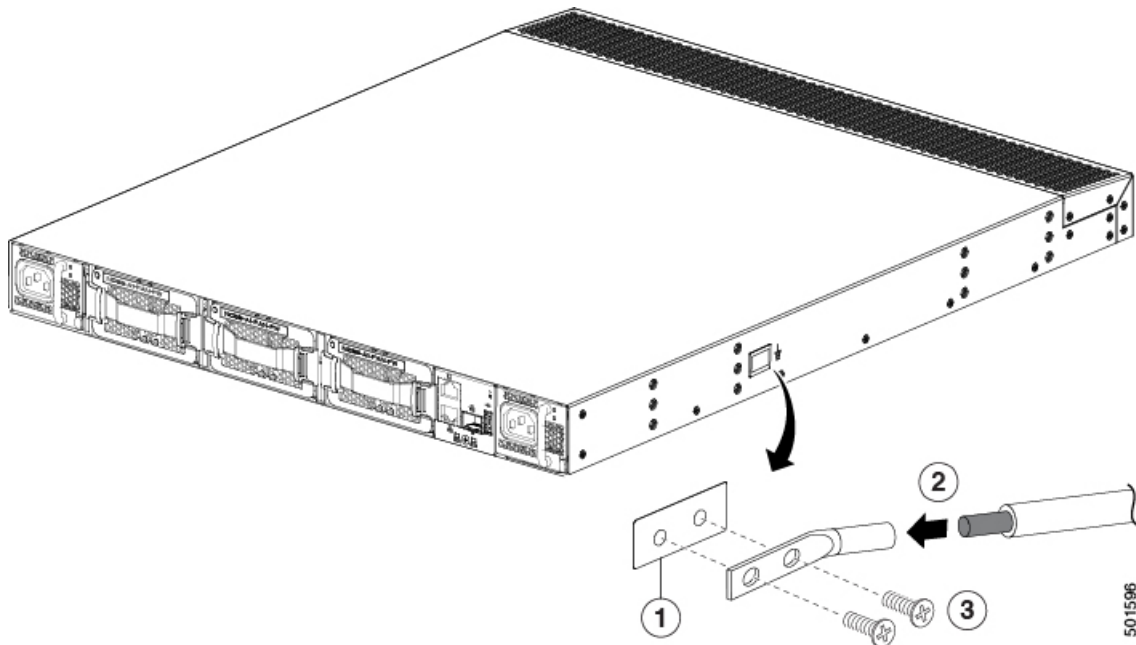
When installing or replacing the unit, the ground connection must always be made first and disconnected last.

Before you begin

Before you can ground the chassis, you must have a connection to the earth ground for the data center building.

- Step 1** Use a wire-stripping tool to remove approximately 0.75 inch (19 mm) of the covering from the end of the grounding wire.
- Step 2** Insert the stripped end of the grounding wire into the open end of the grounding lug, and use a crimping tool to crimp the lug to the wire (see Callout 2 in the following figure). Verify that the ground wire is securely attached to the grounding lug by attempting to pull the wire out of the crimped lug.

Figure 6: Grounding the Switch.



1	Chassis grounding pad	3	2 M4 screws used to secure the grounding lug to the chassis
2	Grounding cable, with 0.75 in. (19 mm) of insulation stripped from one end, inserted into the grounding lug and crimped in place		

- Step 3** Secure the grounding lug to the chassis grounding pad with two M4 screws (see Callouts 1 and 3 in the previous figure), and tighten the screws to 11 to 15 in-lb (1.24 to 1.69 N·m) of torque.
-

Starting the Switch

You start the switch by connecting it to its dedicated power source. If you need $n+1$ redundancy, you must connect each of the power supplies to one or two power sources. If you need $n+n$ redundancy, you must connect each power supply in a switch to a different power source.

Before you begin

- The switch must be installed and secured to a rack or cabinet.
- The switch must be adequately grounded.
- The rack must be close enough to the dedicated power source so that you can connect the switch to the power source by using a designated power cables.
- You have the designated power cables for the power supplies that you are connecting to the dedicated power sources.

**Note**

Depending on the outlet receptacle on your AC power distribution unit, you might need an optional jumper power cord to connect the switch to your outlet receptacle.

- The switch is not connected to the network (this includes any management or interface connections).
- The fan and power supply modules are fully secured in their chassis slots.

All of the fan slots must be filled with fan modules and the power supply slots must be filled with the same types of power supplies (do not mix AC and DC power supplies).

- Step 1** Verify that the power supply  LED is on and green.
- Step 2** Listen for the fans; they should begin operating when the power supply is powered.
- Step 3** After the switch boots, verify that the following LEDs are lit:
- On the fan modules, the Status (STA or STS) LED is green.
- If a fan module Status LED is not green, try reinstalling the fan module.
- After initialization, the switch chassis Status (labeled as STA or STS) LED is green.
- Step 4** Verify that the system software has booted and the switch has initialized without error messages.
- Step 5** Complete the worksheets provided in [Site Preparation and Maintenance Records](#) for future reference.

Note

A setup utility automatically launches the first time that you access the switch and guides you through the basic configuration. For instructions on how to configure the switch and check module connectivity, see the appropriate Cisco Nexus configuration guide.



CHAPTER 4

Connecting the Switch to the ACI Fabric

- [ACI Fabric Topology, on page 23](#)
- [Preparing to Connect to Other Devices, on page 24](#)
- [Connecting Leaf Switches to APICs, on page 25](#)
- [Connecting Leaf Switches to Spine Switches, on page 26](#)
- [Setting Up an Optional Console Interface, on page 29](#)
- [Setting Up an Optional Management Connection, on page 30](#)
- [Maintaining Transceivers and Optical Cables, on page 30](#)

ACI Fabric Topology

The ACI fabric topology includes the following major components:

- Application Centric Infrastructure Controller (APIC) appliance (cluster of APICs)
- Leaf switches (Cisco Nexus 93108TC-EX, 93108TC-FX, 93120TX, 93128TX, 93180LC-EX, 93180YC-EX, 93180YC-FX, 9332PQ, 9336C-FX2, 9348GC-FXP, 9372PX, 9372PX-E, 9372TX, 9372TX-E, 9396PX, and 9396TX switches)
- Spine switches (Cisco Nexus 9336PQ, 9364C, 9504, 9508, and 9516 switches)

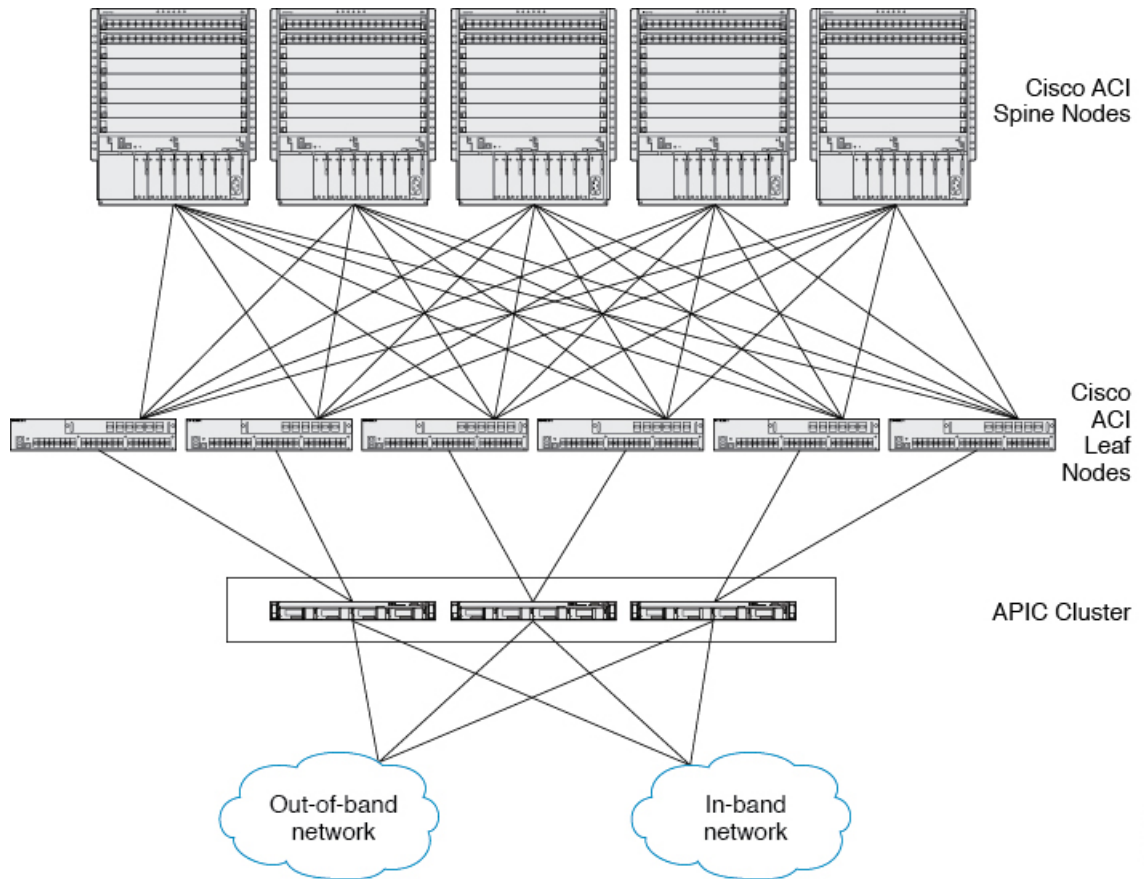
As shown in the following figure, each APIC is connected to one or two leaf switches and each leaf switch should be connected to every spine switch in the same fabric.



Note

To prevent sub-optimal forwarding between endpoints, connect every leaf switch in the fabric to every spine switch in the same fabric.

Figure 7: Connections Between APIC Clusters, Leaf Nodes, and Spine Nodes



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Preparing to Connect to Other Devices

When preparing to connect the fabric devices, consider the following for each type of interface, and gather all of the required equipment before making the connections:

- Cabling type required for each interface type
- Distance limitations for each signal type
- Additional interface equipment required



Note

When running power and data cables in overhead or subfloor cable trays, we strongly recommend that you locate power cables and other potential noise sources as far away as practical from network cabling that terminates on Cisco equipment. In situations where long parallel cable runs cannot be separated by at least 3.3 feet (1 meter), we recommend that you shield any potential noise sources by housing them in a grounded metallic conduit.

The optical transceivers that are not already assembled to their cables come separate from their cables. To prevent these transceivers and their cables from being damaged, we recommend that you keep the transceivers disconnected from their cables when installing them in ports and then insert the optical cable into the transceiver. When removing transceivers from ports, remove their cables before removing the transceivers.

To maximize the effectiveness and life of your transceivers and optical cables, do the following:

- Wear an ESD-preventative wrist strap that is connected to an earth ground whenever handling transceivers. The switch is typically grounded when you install transceivers and provides an ESD port to which you can connect your wrist strap. If you cannot find an ESD port, connect the wrist strap to an earth ground (such as the grounding connection for the chassis).
- Do not remove or insert a transceiver more often than necessary. Repeated removals and insertions can shorten its useful life.
- Keep the transceivers and fiber-optic cables clean and dust free to maintain high signal accuracy and to prevent damage to the connectors. Attenuation (loss of light) increases with contamination and should be kept below 0.35 dB.
 - Clean these parts before installing them to prevent dust from scratching the fiber-optic cable ends.
 - Clean the connectors regularly; the required frequency of cleaning depends upon the environment. In addition, clean connectors if they are exposed to dust or accidentally touched. Both wet and dry cleaning techniques can be effective; refer to your site's fiber-optic connection cleaning procedures.
 - Do not touch the ends of connectors. Touching the ends can leave fingerprints and cause other contamination.
- Inspect routinely for dust and damage. If you suspect damage, clean and then inspect fiber ends under a microscope to determine if damage has occurred.

Connecting Leaf Switches to APICs

You must downlink one or two (recommended for redundancy) ACI-mode leaf switches (Cisco Nexus 93108TC-EX, 93108TC-FX, 93120TX, 93128TX, 93180LC-EX, 93180YC-EX, 93180YC-FX, 9332PQ, 9336C-FX2, 9348GC-FXP, 9372PX, 9372PX-E, 9372TX, 9372TX-E, 9396PX, or 9396TX) to each Application Policy Infrastructure Controller (APIC) in your ACI fabric. The type of virtual interface card (VIC) installed on the APIC determines the types of interface cables that you can use to connect the leaf switches to the APICs.

- The VIC1225 module supports optical transceivers, optical cables, and switches with optical downlink ports (Cisco Nexus 93180LC-EX, 93180YC-EX, 93180YC-FX, 9332PQ, 9336C-FX2, 9348GC-FXP, 9372PX, 9372PX-E, and 9396PX switches).
- The VIC1225T module supports copper connectors, copper cables, and switches with copper downlink ports (Cisco Nexus 93108TC-EX, 93108TC-FX, 93120TX, 93128TX, 9372TX, 9372TX-E, and 9396TX switches).

Before you begin

The APIC and leaf switches in the fabric must be fully installed in their racks and grounded.

Step 1 Connect an interface cable to one of the two ports on the virtual interface card (VIC) installed on the APIC. If the cable is not already assembled to its transceivers, insert the transceiver into the VIC port and then connect the optical interface cable to the transceiver.

- For a VIC1225 optical module, use one of the following sets of transceivers and cables:
 - Cisco 10GBASE-LR transceivers (SFP-10G-LR) supporting a link length of up to 6.1 miles (10 km)
 - Cisco 10GBASE-SR transceivers (SFP-10G-SR) supporting the following link lengths:
 - Using 2000 MHz MMF (OM3) for up to 984 feet (300 m)
 - Using 4700 MHz MMF (OM4) for up to 1312 feet (400 m)
 - Cisco SFP+ Active Optical Cables (SFP-10G-AOCxM [where x=1, 2, 3, 5, 7, or 10 for lengths in meters])

To determine which transceivers, adapters, and cables are supported by this switch, see

<http://www.cisco.com/c/en/us/support/interfaces-modules/transceiver-modules/products-device-support-tables-list.html>.

To see the transceiver specifications and installation information, see

<http://www.cisco.com/c/en/us/support/interfaces-modules/transceiver-modules/products-installation-guides-list.html>.

- For a VIC1225T 10GBASE-T copper module, use 10GBASE-T cables with RJ-45 connectors.

Step 2 Connect the other end of the interface cable to a downlink port on a leaf switch.

- For a Cisco 10GBASE-LR or -SR transceiver and cable, insert the transceiver into a downlink optical port on a leaf switch before connecting the cable to the transceiver.
- For Cisco SFP+ Active Optical Cables, insert the transceiver on the cable into a downlink optical port on a leaf switch.
- For a 10GBASE-T copper cable, insert the RJ-45 connector on the cable into a downlink BASE-T port on a leaf switch.

Note To determine which transceivers and cables are supported by this switch, see <http://www.cisco.com/c/en/us/support/interfaces-modules/transceiver-modules/products-device-support-tables-list.html>.

Connecting Leaf Switches to Spine Switches

For optimal forwarding between endpoints, you must connect each leaf switch (Cisco Nexus 93108TC-EX, 93108TC-FX, 93120TX, 93128TX, 93180LC-EX, 93180YC-EX, 93180YC-FX, 9332PQ, 9336C-FX2, 9348GC-FXP, 9372PX, 9372PX-E, 9372TX, 9372TX-E, 9396PX, or 9396TX) to every spine switch (Cisco Nexus 9336PQ, 9364C, 9504, 9508, or 9516) in the same ACI fabric. The following table lists the number of ports that you can connect on each type of leaf switch and the supported speeds for those ports.

Leaf Switch	Default Uplink Connections	Supported Transmission Speeds (Uplink Ports)
Cisco Nexus 93108TC-EX	6 QSFP28 fixed ports	10, 40, and 100 Gigabits ¹

Leaf Switch		Default Uplink Connections	Supported Transmission Speeds (Uplink Ports)
Cisco Nexus 93108TC-FX		6 QSFP28 fixed ports	10, 40, and 100 Gigabits
Cisco Nexus 93120TX		6 QSFP+ fixed ports	40 Gigabits
Cisco Nexus 93128TX	with M6PQ or M4PQ-E uplink module	6 QSFP+ ports	40 Gigabits
	with M12PQ uplink module	8 QSFP+ ports (leftmost 8 ports are supported on 12-port module)	1, 10, or 40 Gigabits ¹
Cisco Nexus 93180LC-EX		6 QSFP28 fixed ports	40, and 100 Gigabits
Cisco Nexus 93180YC-EX		6 QSFP28 fixed ports	10, 40, and 100 Gigabits ¹
Cisco Nexus 93180YC-FX		6 QSFP28 fixed ports	10, 40, and 100 Gigabits ¹
Cisco Nexus 9332PQ		6 QSFP+ fixed ports	10 or 40 Gigabits ¹
Cisco Nexus 9336C-FX2		6 QSFP28 fixed ports	10, 40, or 100 Gigabits
Cisco Nexus 9348GC-FXP		2 QSFP28 fixed ports	40 or 100 Gigabits
Cisco Nexus 9372PX Cisco Nexus 9372PX-E Cisco Nexus 9372TX Cisco Nexus 9372TX-E		6 QSFP+ ports	40 Gigabits
Cisco Nexus 9396PX Cisco Nexus 9396TX	with M6PQ or M4PQ-E uplink module	6 QSFP+ ports	40 Gigabits
	with M12PQ uplink module	12 QSFP+ ports	10 or 40 Gigabits ¹

¹ For 10-Gigabit support, use a QSFP-to-SFP adapter (such as CVR-QSFP-SFP10G) with a SFP+ or SFP transceiver.

The following table lists the number of ports that you can connect on each ACI-mode line card installed in a modular spine switch (Cisco Nexus 9504, 9508, or 9516 switch) or on each Cisco Nexus 9336PQ spine switch. The number of line cards in a spine switch depends on the model of the spine switch (the Cisco Nexus 9504 supports four line cards, the Cisco Nexus 9508 supports up to eight line cards, and the Cisco Nexus 9516 supports up to 10 line cards in ACI-mode).

Spine Switch or Modular Line Card	Maximum Number of Uplink Connections	Supported Transmission Speeds
Cisco Nexus 9336PQ switch	36 QSFP+ fixed ports	40 Gigabits
N9K-X9732C-EX line card supported by Cisco Nexus modular switches	32 QSFP28 fixed ports	1, 10, 25, 40, 50, or 100 Gigabits

Spine Switch or Modular Line Card	Maximum Number of Uplink Connections	Supported Transmission Speeds
N9K-X9736PQ line card supported by Cisco Nexus 9504, 9508, and 9516 modular switches	36 QSFP+ fixed ports	1, 10, or 40 Gigabits
N9K-X9736C-FX line card supported by Cisco Nexus 9504, 9508, and 9516 modular switches	36 QSFP28 fixed ports	1, 10, 40, or 100 Gigabits

To determine which transceivers, adapters, and cables are supported by this switch, see <http://www.cisco.com/c/en/us/support/interfaces-modules/transceiver-modules/products-device-support-tables-list.html>. To see the transceiver specifications and installation information, see <http://www.cisco.com/c/en/us/support/interfaces-modules/transceiver-modules/products-installation-guides-list.html>.



Warning Statement 1053—Class 1M Laser Radiation

Class 1M laser radiation when open. Do not view directly with optical instruments.



Warning Statement 1055—Class I and Class 1M Laser

Class I (CDRH) and Class 1M (IEC) laser products.



Warning Statement 1056—Unterminated Fiber Cable

Invisible laser radiation may be emitted from the end of the unterminated fiber cable or connector. Do not view directly with optical instruments. Viewing the laser output with certain optical instruments (for example, eye loupes, magnifiers, and microscopes) within a distance of 100 mm may pose an eye hazard.

Before you begin

- The leaf and spine switches in the fabric must be fully installed in their racks and grounded.
- If there are modular switches in the fabric, their ACI-mode line cards must already be installed. The line cards can be of the following types:
 - 36-port 40-Gigabit (N9K-X9736PQ)
 - 32-port 100-Gigabit (N9K-X9732C-EX) (supported by Cisco Nexus 9504 and 9508 modular switches)



Note You cannot include NX-OS line cards in the same chassis when running in ACI mode.

- Step 1** For the transceivers with removable cables, make sure that the transceivers are separated from their interface cables.
- Step 2** Insert the appropriate transceiver into an active uplink port on the leaf switch.

Note For 1- or 10-Gigabit Ethernet on a Cisco Nexus 93108TC-EX or 93180YC-EX switch or a N9K-X9732C-EX, N9K-X9736PQ, or N9K-X9432PQ line card, insert a QSFP-to-SFP adapter (CVR-QSFP-SFP10G) before inserting an SFP+ transceiver.

- Step 3** Insert the same type of transceiver in the spine switch port on a line card.
- Step 4** For transceivers with removable cables, insert the interface cable into the open end of each of those transceivers.
- Step 5** Repeat Steps 1 through 4 for each spine switch in the ACI fabric.
The leaf switch is connected to each spine switch in the ACI fabric.
- Step 6** Repeat Steps 1 through 5 for each leaf switch in the ACI fabric.
Each leaf switch in the ACI fabric is connected to each spine switch in the network,

The fabric automatically implements Equal Cost Multi-Pathing (ECMP) and enables all links. You do not need to configure the links.

Setting Up an Optional Console Interface

You can optionally set up a console interface for performing the initial configuration of the switch. To do this, use the interface cable provided in the accessory kit to connect the switch to your console device. You can connect the console port on the switch to a modem. If you do not connect it to a modem, make the connection either before powering up the switch or after completing the boot process for the switch.

Before you begin

The console device must support VT100 terminal emulations and asynchronous transmissions.

-
- Step 1** Configure the terminal emulator program to match each of the following default port characteristics:
- 9600 baud
 - 8 data bits
 - 1 stop bit
 - No parity
- Step 2** Insert the RJ-45 connector on the interface cable found in the accessory kit into the RS-232 port on the switch and insert the DB-9 connector on the other end of the cable to the serial port on the console device.
-

What to do next

You can now perform the initial configuration for the switch (see the *Cisco ACI Getting Started Guide*).

Setting Up an Optional Management Connection

You can optionally set up an out-of-band management connection for monitoring and troubleshooting purposes. To do this, you connect either the RJ-45 management port or the SFP management port on the switch to an external hub, switch, or router.

Before you begin

To prevent an IP address conflict, you must complete the initial configuration for the switch and establish an IP address before you create the management connection.

- Step 1** Connect the appropriate modular interface cable to only one of the two management ports on the switch. Connect the management interface cable to a management port on the leaf switch. If the switch has two management ports, use the optical port if you have an optical interface cable or use the RJ-45 port if you have a copper interface cable.
- For the RJ-45 management port, use a copper interface cable with RJ-45 connectors (can be used for shorter connections).
 - For the SFP management port, use an optical interface cable with LH or SX SFP transceivers (can be used for longer connections).

Note If you use the management interface, connect only one of the two management ports. The switch does not support your use of two management ports at the same time.

- Step 2** Connect the other end of the cable to an external hub, switch, or router.

Maintaining Transceivers and Optical Cables

Transceivers and fiber-optic cables must be kept clean and dust free to maintain high signal accuracy and prevent damage to the connectors. Attenuation (loss of light) is increased by contamination and should be below 0.35 dB.

Consider the following maintenance guidelines:

- Transceivers are static sensitive. To prevent ESD damage, wear an ESD-preventative wrist strap that is connected to the grounded chassis.
- Do not remove and insert a transceiver more often than is necessary. Repeated removals and insertions can shorten its useful life.
- Keep all optical connections covered when not in use. Clean them before using to prevent dust from scratching the fiber-optic cable ends.
- Do not touch the ends of connectors. Touching the ends can leave fingerprints and cause other contamination.
- Clean the connectors regularly; the required frequency of cleaning depends upon the environment. In addition, clean connectors if they are exposed to dust or accidentally touched. Both wet and dry cleaning techniques can be effective; refer to your site's fiber-optic connection cleaning procedures.

- Inspect routinely for dust and damage. If you suspect damage, clean and then inspect fiber ends under a microscope to determine if damage has occurred.



CHAPTER 5

Replacing Components

- [Replacing a Fan Module, on page 33](#)
- [Replacing a Power Supply Module, on page 35](#)

Replacing a Fan Module

You can replace one of the three fan modules at a time while the switch is operating so long as you perform the replacement within one minute. If you cannot perform the replacement within one minute, leave the original fan module in the chassis to maintain the designed airflow until you have the replacement fan module on hand and can perform the replacement.



Caution

If you are replacing a module during operations, be sure that the replacement fan module has the correct direction of airflow, which means that it has the same airflow direction as the other modules in the chassis. Also, be sure that the airflow direction takes in air from a cold aisle and exhausts to a hot aisle. Otherwise, the switch can overheat and shutdown.

If you are changing the airflow direction of all the modules in the chassis, you must shutdown the switch before replacing all the fan and power supply modules with modules using the other airflow direction. During operations, all of the modules must have the same direction of airflow.

Removing a Fan Module



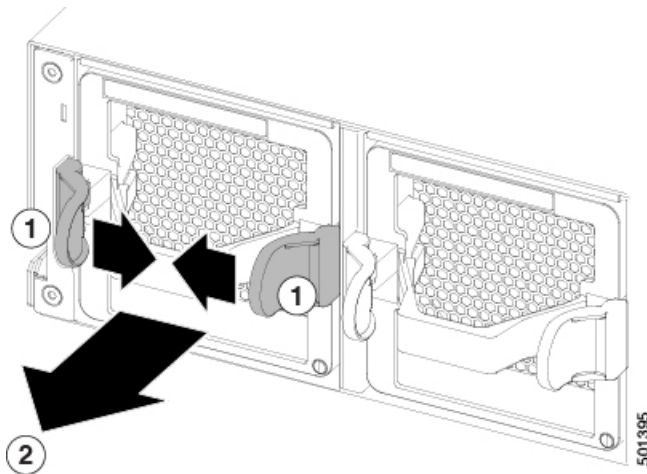
Warning

Statement 263—Fan Warning

The fans might still be turning when you remove the fan assembly from the chassis. Keep fingers, screwdrivers, and other objects away from the openings in the fan assembly's housing.

Step 1

On the fan module that you are removing, press the two colored handles towards each other and pull the module part way out of its slot to unseat it from its connectors.



1	Press the two colored handles towards each other.	2	Pull the pressed handles to remove the module from its slot.
---	---	---	--

Step 2 Holding the handle, pull the module out of the chassis.

Caution Do not touch the electrical connectors on the back side of the module and prevent anything else from coming into contact with and damaging the connectors.

Installing a Fan Module

Before you begin

- A fan slot must be open and ready for the new fan module to be installed.
- You must have a new fan module on hand and ready to install within one minute of removing the original fan module if the switch is operating.
- The new fan module must have the same airflow direction as the other fan and power supply modules installed in the switch. All of these modules must have either burgundy coloring (port-side intake airflow) or they must all have blue coloring (port-side exhaust airflow).

Step 1 Holding the fan module by its handle, align the back of the fan module (the side with the electrical connectors) to the open fan slot in the chassis.

Step 2 Slide the fan module into the slot until it clicks in place.

Step 3 Verify that the Status (STS) LED turns on and becomes green.

Replacing a Power Supply Module

The switch requires two power supplies for redundancy. With one power supply providing the necessary power for operations, you can replace the other power supply during operations so long as the new power supply has the same airflow direction as the other modules in the chassis.

You can replace a power supply with another supported power supply that has the same power source type (AC, DC, HVAC, or HVDC) and the same wattage rating as the other installed power supply. Additionally, the airflow direction of the power supply must match the airflow direction of the installed fan modules. For the airflow direction used by the switch, see the coloring of the fan modules. The following list describes the power supplies supported by this switch.

- NXA-PAC-1100W-PE2 (1100-W, port-side exhaust (blue latch) power supply requiring AC power source providing at least 16 A)
- NXA-PAC-1100W-PI2 (1100-W, port-side intake (burgundy latch) power supply requiring AC power source providing at least 16 A)
- N9K-PHV-1100W-PE (1100-W, port-side exhaust (blue latch) high-voltage AC/DC power supply requiring a high-voltage AC or DC power source)
- N9K-PHV-1100W-PI (1100-W, port-side intake (burgundy latch) high-voltage AC/DC power supply requiring a high-voltage AC or DC power source)
- NXA-PDC-1100W-PE (1100-W, port-side exhaust (blue latch) power supply requiring DC power source providing at least 16 A)
- NXA-PDC-1100W-PI (1100-W, port-side intake (burgundy latch) power supply requiring DC power source providing at least 16 A)

Replacing an AC Power Supply

You can replace an AC power supply during operations so long as the other power supply provides to the switch.

Before you begin

- The replacement power supply must have the same wattage and airflow direction as the power supply being replaced.



Note

You can determine the airflow direction by looking at the coloring of the latch on each power supply. AC power supplies with burgundy latches have port-side intake airflow direction, and power supplies with blue latches have port-side exhaust airflow direction.

- An AC power source must be within reach of the power cable that will be used with the replacement power supply. If you are using $n+n$ power redundancy, there must be a separate power source for each power supply installed in the chassis.

- There must be an earth ground connection to the chassis that you are installing the replacement module. AC power supplies connected to AC power sources are automatically grounded through their power cable.

Step 1 Remove an AC power supply as follows:

- Holding the plug for the power cable, pull the plug out from the power receptacle on the power supply and verify that both power supply LEDs are off.
- Grasp the power supply handle while pressing the colored release latch towards the power supply handle.
- Place your other hand under the power supply to support it while you slide it out of the chassis.

Caution Do not touch the electrical connections on the back side of the module and prevent anything else from coming into contact with and damaging the connectors.

Step 2 Install the replacement power supply as follows:

- Holding the replacement power supply with one hand underneath the module and the other hand holding the handle, turn the power supply so that its release latch is on the right side and align the back end of the power supply (the end with the electrical connections) to the open power supply slot before carefully sliding the power supply all the way into the slot until it clicks into place.

Note If the power supply does not fit into the open slot, turn the module over before sliding it carefully into the open slot.

- Test the installation by trying to pull the power supply out of the slot without using the release latch.

If the power supply does not move out of place, it is secured in the slot. If the power supply moves, carefully press it all the way into the slot until it clicks in place.

- Attach the power cable to the electrical outlet on the front of the power supply.
- Make sure that the other end of the power cable is attached to the appropriate power source for the power supply.

Note Depending on the outlet receptacle on your power distribution unit, you might need the optional jumper cable to connect the switch to your outlet receptacle.

- Verify that the power supply is operational by making sure that the power supply  LED is green. For information on what the power supply LEDs indicate, see [Power Supply LEDs, on page 50](#).

Replacing a High Voltage (HVAC/HVDC) Power Supply

You can replace an HVAC/HVDC power supply during operations so long as the other power supply provides power to the switch.

Before you begin

- The replacement power supply must have the same wattage and airflow direction as the power supply being replaced.



Note You can determine the airflow direction by looking at the coloring of the latch on each power supply. The high voltage power supplies have either burgundy latches for port-side intake airflow or they have blue latches for port-side exhaust airflow.

- An HVAC/HVDC power source must be within reach of the power cable that will be used with the replacement power supply. If you are using $n+n$ power redundancy, there must be a separate power source for each power supply installed in the chassis.
- There must be an earth ground connection to the chassis in which you are installing the replacement power supply. HVAC/HVDC power supplies connected to AC power sources are automatically grounded by their power cable when connected to the power supply and AC power source. HVAC/HVDC power supplies connected to DC power sources have Saf-D-Grid power cables with three connectors on the power source end--you connect one of those connectors to the earth ground.

Step 1 Remove an HVAC/HVDC power supply as follows:

- a) Turn off the circuit breaker for the power feed to the power supply that you are replacing.
Be sure that the LEDs turn off on the power supply that you are removing.
- b) Remove the power cable from the power supply by pressing the tab on the top of the Anderson Power SAF-D-Grid connector and pull the cable and connector out of the power supply.
- c) Grasp the power supply handle while pressing the colored release latch towards the power supply handle.
- d) Place your other hand under the power supply to support it while you slide it out of the chassis.

Caution Do not touch the electrical connections on the back side of the module and prevent anything else from coming into contact with and damaging the connectors.

Step 2 Install the replacement power supply as follows:

- a) Holding the replacement power supply with one hand underneath the module and the other hand holding the handle, turn the power supply so that its release latch is on the right side and align the back end of the power supply (the end with the electrical connections) to the open power supply slot before carefully sliding the power supply all the way into the slot until it clicks into place.

Note If the power supply does not fit into the open slot, turn the module over before sliding it carefully into the open slot.

- b) Test the installation by trying to pull the power supply out of the slot without using the release latch.

If the power supply does not move out of place, it is secured in the slot. If the power supply moves, carefully press it all the way into the slot until it clicks in place.

- c) Attach the Saf-D-Grid end of the power cable to the electrical outlet on the front of the power supply.
- d) Make sure that the other end of the power cable is attached to the appropriate power source for the power supply.
 - For an HVAC power source, plug the other end of the power cable into the power source.
 - For a HVDC power source, verify that the circuit breaker is turned off and then connect each of the three cable connectors to the appropriate DC and grounding terminals on the power source. If there is a cover plate for the DC terminals, install the plate to prevent accidental contact with the terminals.

- e) If using an HVDC power source, turn on the circuit breaker for the power source.
- f) Verify that the power supply is operational by making sure that the power supply  LED is green. For information on what the power supply LEDs indicate, see [Power Supply LEDs, on page 50](#).

Replacing a DC Power Supply

You can replace an DC power supply during operations so long as the other power supply provides power to the switch.

Before you begin

- The replacement power supply must have the same wattage and airflow direction as the power supply being replaced.



Note

You can determine the airflow direction by looking at the coloring of the latch on each power supply. AC power supplies with burgundy latches have port-side intake airflow direction, and power supplies with blue latches have port-side exhaust airflow direction.

- An DC power source must be within reach of the power cables that will be used with the replacement power supply. If you are using $n+n$ power redundancy, there must be a separate power source for each power supply installed in the chassis.
- There must be an earth ground connection to the chassis in which you are installing the replacement power supply. DC power supplies connected to DC power sources have three power cables (two for DC power and one for grounding).

Step 1

Remove a DC power supply as follows:

- a) Turn off the circuit breaker for the power feed to the power supply that you are replacing.
Be sure that the LEDs turn off on the power supply that you are removing.
- b) Remove the DC power connector block from the power supply by doing the following:
 - Push the orange plastic button on the top of the connector block inward toward the power supply.
 - Pull the connector block out of the power supply.
- c) Grasp the power supply handle while pressing the release latch towards the power supply handle.
- d) Place your other hand under the power supply to support it while you slide it out of the chassis.

Caution Do not touch the electrical connections on the back side of the module and prevent anything else from coming into contact with and damaging the connectors.

Step 2

Install the replacement power supply as follows:

- a) Holding the replacement power supply with one hand underneath the module and the other hand holding the handle, turn the power supply so that its release latch is on the right side and align the back end of the power supply (the end with the electrical connections) to the open power supply slot before carefully sliding the power supply all the way into the slot until it clicks into place.

Note If the power supply does not fit into the open slot, turn the module over before sliding it carefully into the open slot.

- b) Test the installation by trying to pull the power supply out of the slot without using the release latch.

If the power supply does not move out of place, it is secured in the slot. If the power supply moves, carefully press it all the way into the slot until it clicks in place.

- c) Attach the power connector block end of the power cable to the electrical outlet on the front of the power supply.
 - d) Turn on the circuit breaker for the power source.
 - e) Verify that the power supply is operational by making sure that the power supply  LED is green. For information on what the power supply LEDs indicate, see [Power Supply LEDs, on page 50](#).
-



APPENDIX **A**

Rack Specifications

- [Overview of Racks, on page 41](#)
- [General Requirements for Cabinets and Racks, on page 41](#)
- [Requirements Specific to Standard Open Racks, on page 42](#)
- [Requirements Specific to Perforated Cabinets, on page 42](#)
- [Cable Management Guidelines, on page 42](#)

Overview of Racks

You can install the switch in the following types of cabinets and racks, assuming an external ambient air temperature range of 0 to 104°F (0 to 40°C):

- Standard perforated cabinets
- Solid-walled cabinets with a roof fan tray (bottom to top cooling)
- Standard open racks



Note If you are selecting an enclosed cabinet, we recommend one of the thermally validated types, either standard perforated or solid-walled with a fan tray.



Note We do not recommend that you use racks that have obstructions (such as power strips), because the obstructions could impair access to field-replaceable units (FRUs).

General Requirements for Cabinets and Racks

The cabinet or rack must also meet the following requirements:

- Standard 19-inch (48.3 cm) (four-post EIA cabinet or rack, with mounting rails that conform to English universal hole spacing per section 1 of ANSI/EIA-310-D-1992). For more information, see [Requirements Specific to Perforated Cabinets, on page 42](#).

- The minimum vertical rack space requirement per chassis is .
- The width between the rack-mounting rails must be at least 17.75 inches (45.0 cm) if the rear of the device is not attached to the rack. For four-post EIA racks, this measurement is the distance between the two front rails.

Four-post EIA cabinets (perforated or solid-walled) must meet the following requirements:

- The minimum spacing for the bend radius for fiber-optic cables should have the front-mounting rails of the cabinet offset from the front door by a minimum of 3 inches (7.6 cm).
- The distance between the outside face of the front mounting rail and the outside face of the back mounting rail should be 23.0 to 30.0 inches (58.4 to 76.2 cm) to allow for rear-bracket installation.

Requirements Specific to Standard Open Racks

If you are mounting the chassis in an open rack (no side panels or doors), ensure that the rack meets the following requirements:

- The minimum vertical rack space per chassis must be .
- The distance between the chassis air vents and any walls should be 2.5 inches (6.4 cm).

Requirements Specific to Perforated Cabinets

A perforated cabinet has perforations in its front and rear doors and side walls. Perforated cabinets must meet the following requirements:

- The front and rear doors must have at least a 60 percent open area perforation pattern, with at least 15 square inches (96.8 square cm) of open area per rack unit of door height.
- The roof should be perforated with at least a 20 percent open area.
- The cabinet floor should be open or perforated to enhance cooling.

The Cisco R Series rack conforms to these requirements.

Cable Management Guidelines

To help with cable management, you might want to allow additional space in the rack above and below the chassis to make it easier to route all of the fiber optic or copper cables through the rack.



APPENDIX B

System Specifications

- [Environmental Specifications, on page 43](#)
- [Switch Dimensions, on page 43](#)
- [Switch and Module Weights and Quantities, on page 44](#)
- [Transceiver and Cable Specifications, on page 44](#)
- [Switch Power Input Requirements, on page 44](#)
- [Power Specifications, on page 45](#)
- [Power Cable Specifications, on page 46](#)
- [Regulatory Standards Compliance Specifications, on page 47](#)

Environmental Specifications

Environment		Specification
Temperature	Ambient operating temperature	32 to 104°F (0 to 40°C)
	Ambient non operating	–40 to 158°F (–40 to 70°C)
Relative humidity	Operating	8 to 80% with humidity gradation of 10% per hour
	Non operating	5 to 95%
Altitude	Operating	0 to 13,123 feet (0 to 4,000 meters)

Switch Dimensions

Switch	Width	Depth with Handles	Height
Cisco Nexus 9336C-FX2	17.3 inches (43.9 cm)	24.5 inches (62.3 cm)	1.72 inches (4.4 cm) (1 RU)

Switch and Module Weights and Quantities

Component	Weight per Unit	Quantity
Cisco Nexus 9336C-FX2 Chassis (N9K-C9336C-FX2)	18.8 lb (8.5 kg)	1
Fan Module	—	3
– Port-side exhaust (blue) (NXA-FAN-65CFM-PE)	0.6 lb (0.3 kg)	
– Port-side intake (burgundy) (NXA-FAN-65CFM-PI)	0.6 lb (0.3 kg)	
Power Supply module	—	2 (1 for operations and 1 for redundancy)
– 1100-W AC port-side exhaust (blue) (NXA-PAC-1100W-PE2)	2.42 lb (1.1 kg)	
– 1100-W AC port-side intake (burgundy) (NXA-PAC-1100W-PI2)	2.42 lb (1.1 kg)	
– 1100-W HVAC/HVDC port-side exhaust (blue) (NXA-PUV-1100W-PE)	2.42 lb (1.1 kg)	
– 1100-W HVAC/HVDC port-side intake (burgundy) (NXA-PUV-1100W-PI)	2.42 lb (1.1 kg)	

Transceiver and Cable Specifications

To determine which transceivers, adapters, and cables are supported by this switch, see <https://www.cisco.com/c/en/us/support/interfaces-modules/transceiver-modules/products-device-support-tables-list.html>.

To see the transceiver specifications and installation information, see <https://www.cisco.com/c/en/us/support/interfaces-modules/transceiver-modules/products-device-support-tables-list.html>.

Switch Power Input Requirements

The following table lists the typical amount of power consumed by the switch, and it lists the maximum amount of power that you must provision from the power source as input to the power supply and switch for peak conditions.

**Note**

Some power supplies might have UL listed capabilities that are greater than the maximum power requirements for a switch. To determine the power consumption characteristics for the switch, use the typical and maximum requirements listed in the following table.

Switch	Typical Power Consumption (AC or DC)	Maximum Power Consumption (AC or DC)
Cisco Nexus 9336C-FX2	484 W	840 W

Power Specifications

Power specifications include the specifications for each type of power supply module.

1100-W AC Power Supply Specifications

These specifications apply to the following power supplies:

- NXA-PAC-1100W-PE
- NXA-PAC-1100W-PI

Characteristic	Specification
AC input voltage	Nominal range: 100 and 240 VAC (Range: 90-132 VAC, 180-264 VAC)
AC input frequency	Nominal range: 50 to 60 Hz (Range: 47-63 Hz)
Maximum AC input current	12 A at 115 VAC 6 A at 240 VAC
Maximum input volt-amperes	760 A at 100 VAC
Maximum output power per power supply	1100 W
Maximum inrush current	33 A (sub-cycle duration)
Maximum hold-up time	12 ms at 1100 W
Power supply output voltage	12 VDC
Power supply standby voltage	12 VDC
Efficiency rating	Climate Savers Platinum Efficiency (80Plus Platinum certified)
Form factor	RSP1

Power Cable Specifications

The following sections specify the power cables that you should order and use with AC, HVAC/HVDC, and DC power supplies.

Power Cable Specifications for AC Power Supplies

Locale	Power Cord Part Number	Cord Set Description
	CAB-C13-C14-2M	Power Cord Jumper, C13-C14 Connectors, 6.6 feet (2.0 m)
	CAB-C13-C14-AC	Power cord, C13 to C14 (recessed receptacle), 10 A, 9.8 feet (3 m)
	CAB-C13-CBN	Cabinet jumper power cord, 250 VAC, 10 A, C14-C13 connectors, 2.3 feet (0.7 m)
Argentina	CAB-250V-10A-AR	250 V, 10 A, 8.2 feet (2.5 m)
Australia	CAB-9K10A-AU	250 VAC, 10 A, 3112 plug, 8.2 feet (2.5 m)
Brazil	CAB-250V-10A-BR	250 V, 10 A, 6.9 feet (2.1 m)
European Union	CAB-9K10A-EU	250 VAC, 10 A, CEE 7/7 plug, 8.2 feet (2.5 m)
India	CAB-IND-10A	10 A, 8.2 feet (2.5 m)
Israel	CAB-250V-10A-IS	250 V, 10 A, 8.2 feet (2.5 m)
Italy	CAB-9K10A-IT	250 VAC, 10 A, CEI 23-16/VII plug, 8.2 feet (2.5 m)
North America	CAB-9K12A-NA	125 VAC, 13 A, NEMA 5-15 plug, 8.2 feet (2.5 m)
North America	CAB-AC-L620-C13	NEMA L6-20-C13, 6.6 feet (2.0 m)
North America	CAB-N5K6A-NA	200/240V, 6A, 8.2 feet (2.5 m)
Peoples Republic of China	CAB-250V-10A-CN	250 V, 10 A, 8.2 feet (2.5 m)
South Africa	CAB-250V-10A-ID	250 V, 10 A, 8.2 feet (2.5 m)
Switzerland	CAB-9K10A-SW	250 VAC, 10 A, MP232 plug, 8.2 feet (2.5 m)
United Kingdom	CAB-9K10A-UK	250 VAC, 10 A, BS1363 plug (13 A fuse), 8.2 (2.5 m)

Locale	Power Cord Part Number	Cord Set Description
All except Argentina, Brazil, and Japan	NO-POWER-CORD	No power cord included with switch

HVAC/HVDC Power Cables Supported by ACI-Mode and NX-OS Mode Switches

Power Type	Power Cord Part Number	Cord Set Description
HVAC	CAB-HVAC-SD-0.6M	2-foot (0.6 m) cable with Saf-D-Grid and SD connectors
HVAC	CAB-HVAC-C14-2M	6.6-foot (2.0 m) cable with Saf-D-Grid and C14 connector (use for up to 240 V)
HVAC	CAB-HVAC-RT-0.6M	2-foot (0.6 m) cable with Saf-D-Grid and RT connector
HVDC	CAB-HVDC-3T-2M	6.6-foot (2.0 m) cable with Saf-D-Grid and three terminal connectors
All except Argentina, Brazil, and Japan	NO-POWER-CORD	No power cord included with switch

Regulatory Standards Compliance Specifications

The following table lists the regulatory standards compliance for the switch.

Table 1: Regulatory Standards Compliance: Safety and EMC

Specification	Description
Regulatory compliance	Products should comply with CE Markings according to directives 2004/108/EC and 2006/95/EC.
Safety	• UL 60950-1 Second Edition • CAN/CSA-C22.2 No. 60950-1 Second Edition • EN 60950-1 Second Edition • IEC 60950-1 Second Edition • AS/NZS 60950-1 • GB4943

Specification	Description
EMC: Emissions	• 47CFR Part 15 (CFR 47) Class A • AS/NZS CISPR22 Class A • CISPR22 Class A • EN55022 Class A • ICES003 Class A • VCCI Class A • EN61000-3-2 • EN61000-3-3 • KN22 Class A • CNS13438 Class A
EMC: Immunity	• EN55024 • CISPR24 • EN300386 • KN 61000-4 series
RoHS	The product is RoH-6 compliant with exceptions for leaded-ball grid-array (BGA) balls and lead press-fit connectors.



APPENDIX C

LEDs

- [Switch Chassis LEDs, on page 49](#)
- [Lane Link LEDs, on page 50](#)
- [Fan Module LEDs, on page 50](#)
- [Power Supply LEDs, on page 50](#)

Switch Chassis LEDs

The BCN, STS, and ENV, LEDs are located on the left side of the front of the switch. The port LEDs are appear as triangles pointing up or down to the nearest port.

LED	Color	Status
BCN	Flashing blue	The operator has activated this LED to identify this switch in the chassis.
	Off	This switch is not being identified.
STS	Green	The switch is operational.
	Flashing amber	The switch is booting up.
	Amber	Temperature exceeds minor alarm threshold.
	Red	Temperature exceeds major alarm threshold.
	Off	The switch is not receiving power.
ENV	Green	Fans and power supply modules are operational.
	Amber	At least one fan or power supply module is not operating.
(port)	Green	Port admin state is 'Enabled', SFP is present and the interface is connected (i.e. cabled, and the link is up).
	Amber	Port admin state is 'Disabled', or the SFP is absent, or both.
	Off	Port admin state is 'Enabled' and SFP is present, but interface is not connected.

Lane Link LEDs

The Lane Link LEDs indicate which of the break out lanes are being checked. If none of the Lane Link LEDs is lit, all four lanes are being checked.

LED Lit	Description
1	Checking lane 1.
2	Checking lane 2.
3	Checking lane 3.
4	Checking lane 4.
None	Checking all four lanes

Fan Module LEDs

The fan module LED is located below the air holes on the front of the module.

LED	Color	Status
STS	Green	The fan module is operational.
	Red	The fan module is not operational (fan is probably not functional).
	Off	Fan module is not receiving power.

Power Supply LEDs

The power supply LEDs are located on the left front portion of the power supply. Combinations of states indicated by the Okay (🟢) and Fault (⚠️) LEDs indicate the status for the module as shown in the following table.

🟢 LED	⚠️ LED	Status
Green	Off	Power supply is on and outputting power to the switch.
Flashing green	Off	Power supply is connected to a power source but not outputting power to the switch—power supply might not be installed in the chassis.
Off	Off	Power supply is not receiving power or is in shut mode.

 LED	 LED	Status
Green	Flashing amber	Power supply warning—possibly one of the following conditions: • High voltage • High power • Low voltage • Power supply warning condition • Slow power supply fan
Green	Amber	Power supply failure



APPENDIX D

Additional Kits

- [Accessory Kit, on page 53](#)

Accessory Kit

The following table lists and illustrates the contents for the accessory kit (N3K-C3064-ACC-KIT).

Illustration	Description	Quantity
	Mount kit 1/2-mount angled bracket 1/2-mount slider bracket (2) U-mount rails (2) 1/2 x 7 mm mounting screws (16)	1
	Cable with DB-9F and RJ45 connectors	1
	Ground lug kit • Two-hole lug (1) • M4 x 8-mm Phillips pan-head screws (2)	1 kit
Not applicable	EAC Compliance document	1
Not applicable	Hazardous substances list for customers in China	1

**Note**

If you do not receive a part listed in this document, contact Cisco Technical Support at this URL:
<https://www.cisco.com/warp/public/687/Directory/DirTAC.shtml>.

If you purchased this product through a Cisco reseller, you might receive additional contents in your kit, such as documentation, hardware, and power cables.

For a list of the available power cables, see [Power Cable Specifications, on page 46](#).



APPENDIX E

Site Preparation and Maintenance Records

- [Site Preparation Checklist, on page 55](#)
- [Contact and Site Information, on page 56](#)
- [Chassis and Module Information, on page 57](#)

Site Preparation Checklist

Planning the location and layout of your equipment rack or cabinet is essential for successful switch operation, ventilation, and accessibility.

The following table lists the site planning tasks that we recommend that you complete before you install the switch. Your completion of each task ensures a successful switch installation.

Planning Activity	Verification Time and Date
Space evaluation:	
Space and layout	
Floor covering	
Impact and vibration	
Lighting	
Physical access	
Maintenance access	
Environmental evaluation:	
Ambient temperature	
Humidity	
Altitude	
Atmospheric contamination	
Airflow	

Planning Activity	Verification Time and Date
Power evaluation:	
Input power type	
Power receptacles	
Receptacle proximity to the equipment	
Dedicated (separate) circuits for power redundancy	
UPS for power failures	
Grounding: proper wire gauge and lugs	
Circuit breaker size	
Grounding evaluation:	
Data center ground	
Cable and interface equipment evaluation:	
Cable type	
Connector type	
Cable distance limitations	
Interface equipment (transceivers)	
EMI evaluation:	
Distance limitations for signaling	
Site wiring	
RFI levels	

Contact and Site Information

Use the following worksheet to record contact and site information for the installation.

Contact person	
Contact phone	

Contact e-mail	
Building/site name	
Data center location	
Floor location	
Address (line 1)	
Address (line 2)	
City	
State/Province	
Contact person	
ZIP/postal code	
Country	

Chassis and Module Information

Use the following worksheet to record information about the switch.

Contract number	
Chassis serial number	
Product number	

Use the following worksheet to record network-related information.

Switch IP address	
Switch IP netmask	
Hostname	
Domain name	
IP broadcast address	
Gateway/router address	
DNS address	

Use the following worksheet to record information about the modules in the switch.

Module Slot	Module Type	Module Serial Number	Notes
Fan module 1			

Module Slot	Module Type	Module Serial Number	Notes
Fan module 2			
Fan module 3			
Power Supply 1			
Power Supply 2			